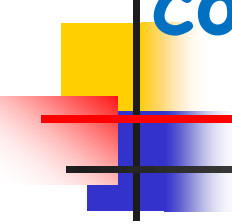


Classification of mycosis

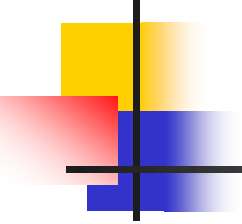
Assist.prof. Dr. Vian B.Barzinji
Medical Mycology



Fungal infections (mycoses) are more common today



- People are living longer (old age)
 - Widespread use of antibiotics
 - Stem cell ,organ transplants and increased immunocompromised person.
-
- **Most mycoses are difficult to treat.** Because fungi are eukaryotes, they share numerous homologous genes and gene products with their human hosts.



Source of infection:

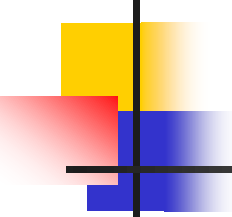
Endogenous:

Normal flora and it is the main source in nosocomial infection (because those people in hospitals are immunocompromised).

Exogenous: This is the main source of fungal infection mainly from the environment.

Fungal Pathogenicity (virulence factors):

- Ability to adhere to host cells by way of cell wall glycoproteins.
- Production capsules allowing them to resist phagocytosis.
- Production of a cytokine called **GM-CSF** by *Candida albicans* that suppress the production of complement.
- Ability to acquire iron from red blood cells as in *Candida albicans*.
- Ability to damage host by secreting enzymes such as **keratinase, elastase, collagenase**.
- Ability to resist killing by phagocytes as in dimorphic fungi(exhibiting **thermal dimorphism**).
- Ability to secrete **mycotoxins**.
- Ability to block the cell-mediated immune defense's of the host.



Host defense factors against fungi

- Physical barriers, such as skin and mucus membranes.
- The fatty acid content of the skin.
- The pH of the skin, mucosal surfaces and body fluids.
- Epithelial cell turnover.
- Normal flora.
- Chemical barriers, such as secretions.
- Most fungi are mesophilic and cannot grow at 37oC.
- Natural Effector Cells (polymorphonuclear leucocytes) and the Professional Phagocytes (monocytes and macrophages).

Types of fungous disease in man(Mycoses)

- A. Allergic compounds of the spore mycelia or other allergins which usually cause sensitization to specific Ag of the fungus. Commonly results in **hypersensitivity** of the individual ends with rhinitis, even asthma.
- B. Toxic compounds produced by some fungi which elaborate directly or indirectly effect that have a toxic effect(**mycotoxin**).
- C. Mycotic (**mycosis**) infection:

A-Hypersensitivity (allergy)

- As a result of dermatophyte infection some fungus-free skin lesions of variable morphology occur elsewhere on the body, which are thought to result from hypersensitivity to the fungus. These reactions are called "**id reaction**".
- These reactions are also seen in *Candida* infections. An inflamed boggy lesion of the scalp called the **kerion** may result from a strong immune reaction to the dermatophyte.
- Many fungi are significant allergens to humans, the allergens being spores, conidia, hyphae and other fungal products. On inhalation they may produce allergic pulmonary diseases such as allergic bronchopulmonary aspergillosis, farmer's lung, bronchial asthma.

B-MYCOTOXINS

- Secondary metabolites produced by food-borne filamentous fungi.
- Vary in their severity :Carcinogen/allergen.
- Non-volatile & low molecular weight.
- Some are lethal.
- Cause identifiable diseases.
- Weaken the immune system without producing symptoms.



ROUTES OF EXPOSURE

- Eating food or drink containing toxins
- Breathing moldy air in damp indoor areas
- Dermal absorption
- Worldwide, approximately 25% of crops are affected by mycotoxins annually
- Mould growing on wooden board



Route of infection

Ingestion/skin contact/ in halation



Blood stream & lymphatic system



Inhibit protein synthesis



Damage macrophages system



Inhibit particle clearance of the lungs



Increase sensitivity to bacterial endotoxins

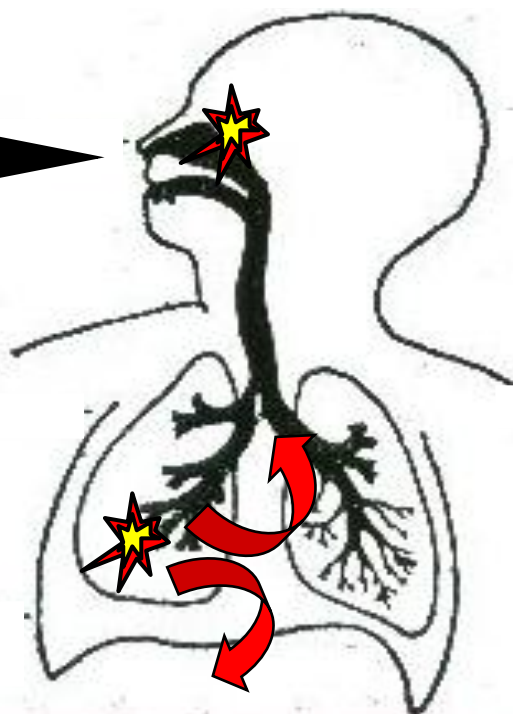
Inhalation of Fungal Spores

DISEASE SPECTRUM

Spores
8 μm



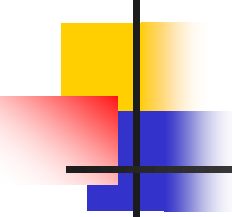
2-4 μm



ALLERGY
SINUSITIS

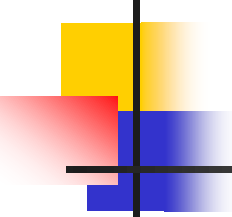
PULMONARY
MYCOSES

DISEMINATED
MYCOSES



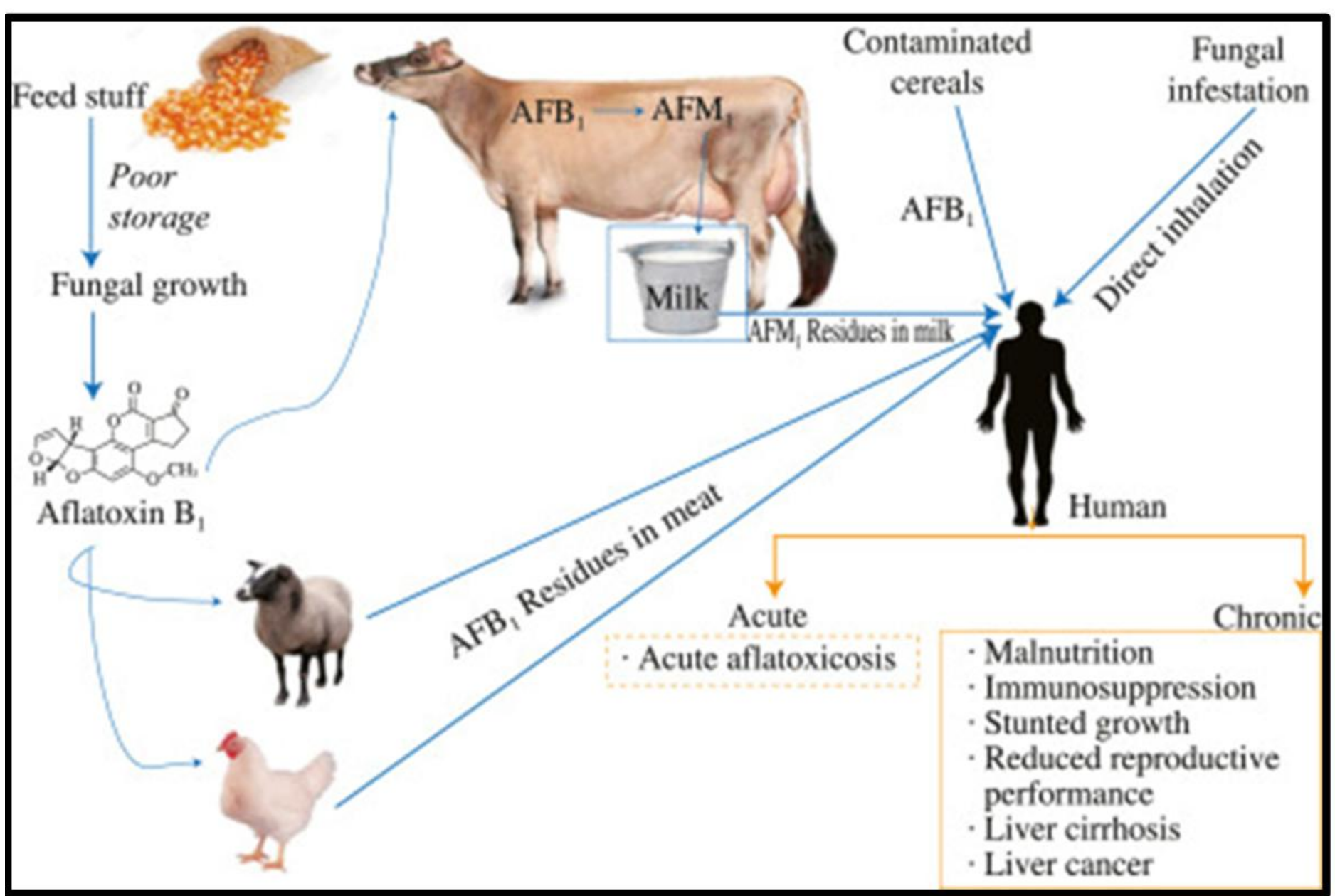
TOXICITY 4 TYPES:

- **Acute:** Rapid onset from single exposure
- **Chronic:** Delayed onset from multiple long term exposure
- **Mutagenic:** Causing damage to DNA
- **Teratogenic:** Causing birth defects



MYCOTOXIN CAN CAUSE:

- Respiratory problems
- Reproductive problems
- Liver, kidney or other organ damage
- Cancer
- Death



Different sources of contamination of aflatoxins to humans

Mycoses classified according to the site of infection

Deep mycoses

Brain

Lungs

Heart

Liver

Spleen

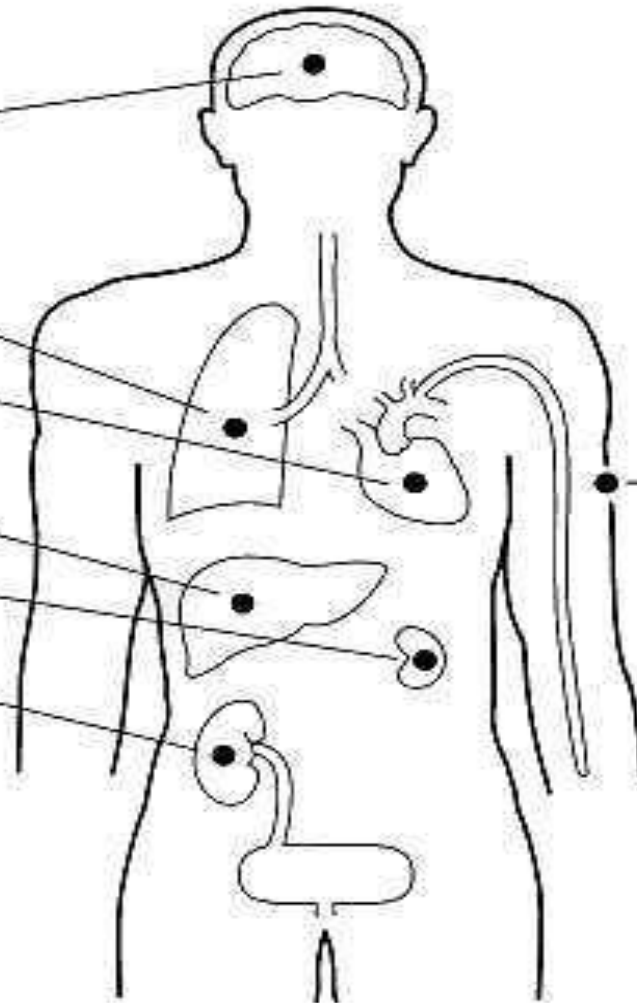
Kidney

Superficial, cutaneous, subcutaneous mycoses

Superficial
(hair, nail, skin)

Cutaneous
(hair, nail, skin)

Subcutaneous



C-Mycoses classified according to the site of infection to:

1-Superficial mycoses:

A-SURFACE MYCOSIS: Live exclusively on **dead surfaces** of skin and its appendages

No contact with living tissue, hence no inflammatory response

1. Tinea versicolor
2. Tinea nigra
3. Piedra



Superficial: B- Cutaneous mycoses

- **Cornified layer** of skin & its appendages
- Contact with living tissue, hence inflammatory & allergic responses seen
 1. Dermatophytes - skin, hair & nails
 2. 3 genera - *Trichophyton*
Microsporum
Epidermophyton



2-Deep mycoses:

Two types: subcutaneous & systemic mycoses

-Caused by soil saprophytes

-Infection is accidental

-Range from a symptomatic infection to fatal disease

A-Subcutaneous mycoses

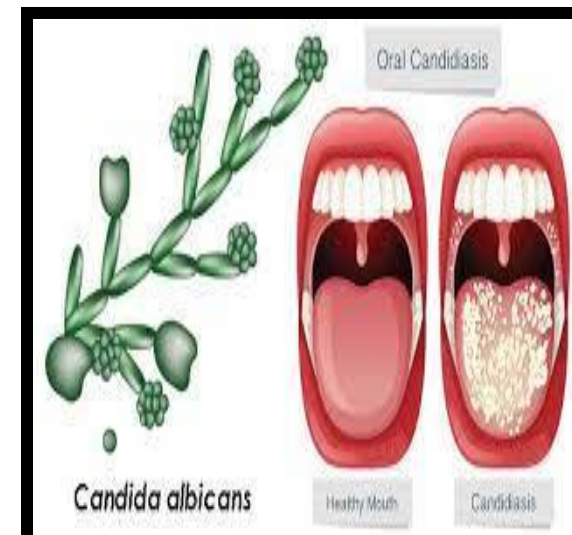
1. Mycotic Mycetoma
2. Chromoblastomycoses
3. Sporotrichosis
4. Rhinosporidiosis
5. Subcutaneous hyphomycoses

B-Systemic mycoses

1. Cryptococcoses
2. Blastomycosis
3. Paracoccidiomycoses
4. Coccidioidomycoses
5. Histoplasmoses

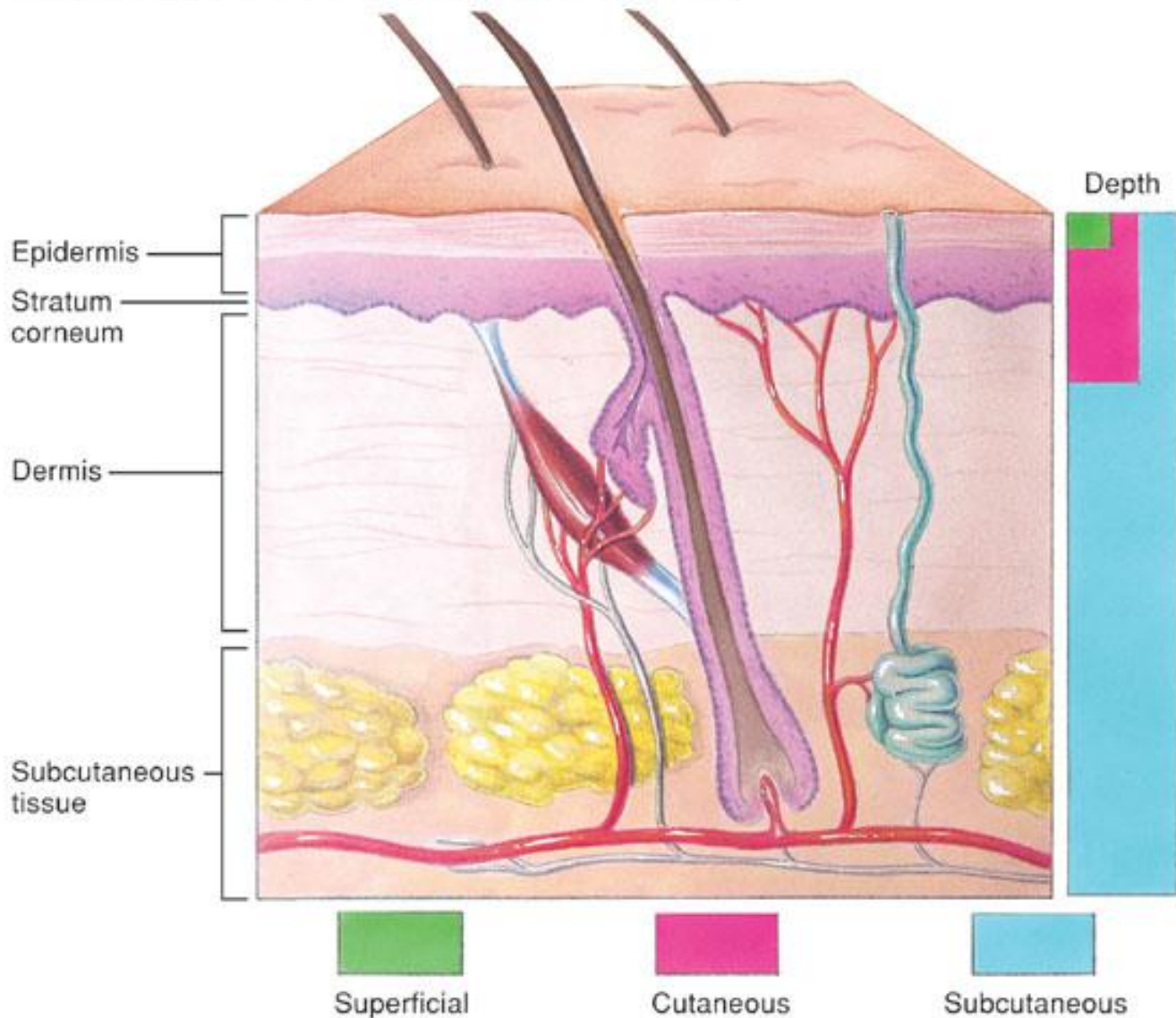
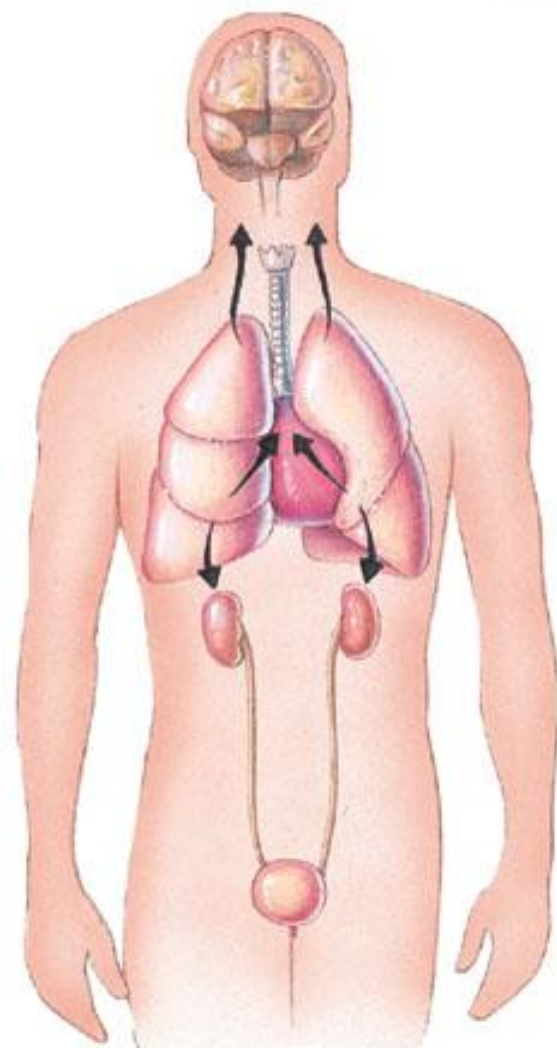
3-Opportunistic infections

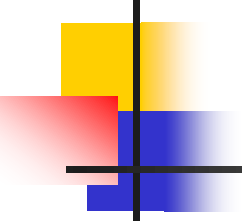
- Opportunistic mycoses are fungal infections that do not normally cause disease in healthy people, but do cause disease in people with weakened immune defenses (immunocompromised people).
- Mainly caused by fungi which are common lab contaminant on culture media
 - *Aspergillus*
 - *Pencillium*
 - *Mucor*
 - *Rhizopus*
- Produce serious & fatal infections



Levels of invasion

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THANK YOU FOR
LISTENING ☺

**GOOD
LUCK!**

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